BRAC University **SET A**

Department of Computer Science and Engineering

CSE111: Programming Language II

Examination: *Quiz #3 (Re)* Semester: *Summer 2022*
Date: 24 / 08 / 2022 Time: 30 Minutes

ID:	Name:	Obtained Points:	Section:
_ _ _ _ _ _ _ _	*(Please write in **CAPITAL LETTERS**)*	**/ 20**	**42**

- Read questions carefully.
- Understanding the question is part of the exam, please do not ask questions.

1. Subtasks:
   a. Create a Planet class with 2 class variables, 1 class list, and 1 classmethod. (**8 Marks**)
   b. All the instance variables should be strongly private. (**2 Marks**)
   c. Calculation of planet asset is given below: (**8 Marks**)

If mass < 1000 Yg	If mass < 5000 Yg	Otherwise
Planet_asset = mass * 300 / Y	Planet_asset = mass * 350 / Y	Planet_asset = mass * 400 / Y

   d. Calculation of planet's asset percentage = planet_asset / total_asset * 100 (**2 Marks**)

Driver code	Expected output

`Planet.details()`
`print('########################')`
`earth = Planet("Earth")`
`earth.setMass(970)`
`earth.info()`
`print('------------------------')`
`Planet.details()`
 	
`print('========================')`
 	
`mercury = Planet("Mercury")`
`venus = Planet("Venus")`
`venus.setMass(4870)`
`mercury.setMass(330)`
`earth.setMass(5970)`
`Planet.details()`
	Total number of planet(s): 0
	Total asset: $0.0
	########################
	{ ID: 1, Name: Earth, Mass: 970 Yg, Asset: $291000.0 }
	------------------------ 	
	Total number of planet(s): 1
	Total asset: $291000.0
	{ ID: 1, Name: Earth, Mass: 970 Yg, Asset: $291000.0 }
	{{{ Planet consists of total asset's: 100.0% }}}
	========================
	Total number of planet(s): 3
	Total asset: $4191500.0
	{ ID: 1, Name: Earth, Mass: 5970 Yg, Asset: $2388000.0 }
	{{{ Planet consists of total asset's: 56.97% }}}
	{ ID: 2, Name: Mercury, Mass: 330 Yg, Asset: $99000.0 }
	{{{ Planet consists of total asset's: 2.36% }}}
	{ ID: 3, Name: Venus, Mass: 4870 Yg, Asset: $1704500.0 }
	{{{ Planet consists of total asset's: 40.67% }}}

Bonus

2. Trace the following code. **(5 Marks)**

```python
methods = {1: int, 0: str}
result = "0"
a, b = 1, 10
x = 15
i = 0
while i < 8:
    j = 0
    while j < 30:
        q = a + b - (4 + a % 3) / 4
        if int(q) % 10 < 5:
            b = b + int(q)
        else:
            a = a + int(q)
        k = b % a % 3
        method = methods[k]
        result = method(result) + method(x)
        print(result)
        j += b
    i += 1
```

BRAC University **SET B**

Department of Computer Science and Engineering

CSE111: Programming Language II

Examination: *Quiz #3 (Re)* Semester: *Summer 2022*

Date: 24 / 08 / 2022 Time: 30 Minutes

ID:	Name: (Please write in ***CAPITAL LETTERS***)	Obtained Points: / 20	Section: **42**

- Read questions carefully.
- Understanding the question is part of the exam, please do not ask questions.

1. Subtasks:
   a. Create an Employee class. (**2 Marks**)
   b. Create 1 class variable, 1 class list, 1 classmethod, & 1 staticmethod. (**4 Marks**)
   c. All the instance variables should be strongly private. (**4 Marks**)
   d. Calculation of salary is given below: (**8 Marks**)

0 <= working_hours < 5	5 <= working_hours < 7	7 <= working_hours
Salary = working_hours * 400	Salary = working_hours * 350	Salary = working_hours * 300

   e. Calculation of employee’s salary percentage = employee_salary / total_salary * 100 (**2 Marks**)

Driver code

```
Employee.flush()
print('########################')
robert = Employee("RoBERTa", 8)
print(robert.getString())
print('------------------------')
Employee.flush()
print("========================")
james = Employee("James", 3)
albert = Employee("Albert", 12)
robert.setName("Robert")
robert.setWorkingHours(10)
Employee.flush()
print("========================")
print(Employee.calculate_salary(15))
```

Expected output

```
Total employees: 0
Total salary: $0.0
########################
< ID: 1, Name: RoBERTa, WH: 8 hours, Salary: $2400.00 >
------------------------
Total employees: 1
Total salary: $2400.0
< ID: 1, Name: RoBERTa, WH: 8 hours, Salary: $2400.00 >
<<< Employee receives 100.00% salary of total salary >>>
========================
Total employees: 3
Total salary: $7800.0
< ID: 1, Name: Robert, WH: 10 hours, Salary: $3000.00 >
<<< Employee receives 38.46% salary of total salary >>>
< ID: 2, Name: James, WH: 3 hours, Salary: $1200.00 >
<<< Employee receives 15.38% salary of total salary >>>
< ID: 3, Name: Albert, WH: 12 hours, Salary: $3600.00 >
<<< Employee receives 46.15% salary of total salary >>>
========================
4500
```

Bonus

2. Trace the following code. **(5 Marks)**

```
methods = {1: int, 0: str}
result = "0"
a, b = 1, 10
x = 10
i = 0
while i < 8:
    j = 0
    while j < 30:
        q = a + b - (4 + a % 3) / 4
        if int(q) % 10 < 5:
            b = b + int(q)
        else:
            a = a + int(q)
        k = b % a % 3
        method = methods[k]
        result = method(result) + method(x)
        print(result)
        j += b
    i += 1
```

BRAC University — SET C

Department of Computer Science and Engineering

CSE111: Programming Language II

Examination: *Quiz #3 (Re)* — Semester: *Summer 2022*
Date: 24 / 08 / 2022 — Time: 30 Minutes

ID:	Name:	Obtained Points:	Section:
_ _ _ _ _ _ _ _	(*Please write in **CAPITAL LETTERS***)	/ 20	42

- Read questions carefully.
- Understanding the question is part of the exam, please do not ask questions.

1. Subtasks:
   a. Create a Country class with 2 class variables, 1 class list, and 1 classmethod. (**8 Marks**)
   b. All the instance variables should be strongly private. (**2 Marks**)
   c. Calculation of country asset is given below: (**8 Marks**)

0 <= population < 100M	100M <= population < 150M	150M <= population
Country_asset = population * 300 / M	Country_asset = population * 400 / M	Country_asset = population * 350 / M

   d. Calculation of country's asset percentage = *country_asset / total_asset * 100* (**2 Marks**)

Driver code	Expected output

`Country.details()`
`print('########################')`
`bangladesh =`
`Country("Bangladesh")`
`bangladesh.setPopulation(64)`
`bangladesh.info()`
`print('------------------------')`
`Country.details()`
 	
`print('========================')`
 	
`china = Country("China")`
`japan = Country("Japan")`
`japan.setPopulation(125)`
`china.setPopulation(1402)`
`bangladesh.setPopulation(164)`
`Country.details()`
	Total number of country(ies): 0
	Total assets: $0
	########################
	(ID: 0, Name: Bangladesh, Population: 64M, Asset: $19200.0)
	------------------------ 	
	Total number of country(ies): 1
	Total assets: $19200.0
	(ID: 0, Name: Bangladesh, Population: 64M, Asset: $19200.0)
	(((Country consists of total asset's: 100.0%)))
	========================
	Total number of country(ies): 3
	Total assets: $598100.0
	(ID: 0, Name: Bangladesh, Population: 164M, Asset: $57400.0)
	(((Country consists of total asset's: 9.6%)))
	(ID: 1, Name: China, Population: 1402M, Asset: $490700.0)
	(((Country consists of total asset's: 82.04%)))
	(ID: 2, Name: Japan, Population: 125M, Asset: $50000.0)
	(((Country consists of total asset's: 8.36%)))

Bonus

2. Trace the following code. **(5 Marks)**

```
methods = {1: int, 0: str}
result = "0"
a, b = 1, 10
x = 10
i = 0
while i < 8:
    j = 0
    while j < 30:
        q = a + b - (4 + a % 3) / 4
        if int(q) % 10 < 5:
            b = b + int(q)
        else:
            a = a + int(q)
        k = b % a % 3
        method = methods[k]
        result = method(result) + method(x)
        print(result)
        j += b
    i += 1
```